

Bioprocess – *Yeast Fermentation*

A MINI PROJECT REPORT



Submitted by

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in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

In

BIOTECHNOLOGY

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BONAFIDE CERTIFICATE

Certified that this mini project report entitled "**Yeast fermentation**" is the bonafide work of "**THANIKADEVI V (922525214029) & YOGLAKSHMI S (922525214030)**" who carried out the project work under my supervision during the academic year 2025 - 2026.

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EXTERNAL EXAMINER

DECLARATION

We declare that the mini project report entitled "**Yeast Fermentation**" submitted by us for the II Year Mini Project in Bachelor of Technology in Biotechnology is the original and independent work carried out in our college under the guidance of Ms. J. Shanmugapriya, Assistant Professor, Department of Biotechnology, V.S.B. Engineering College, Karur. This document has not previously formed the basis for the award of any degree, diploma, fellowship, or similar academic title.

Date:

Place: Karur

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Thanikadevi V

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DEPARTMENT OF BIOTECHNOLOGY

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We endeavour to impart futuristic technical education of the highest quality to the student community and to inculcate discipline in them to face the world with self-confidence and thus we prepare them for life as responsible citizens to uphold human values and to be of service at large. We strive to bring of the Institution as an Institution of academic excellence of international standard.

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- ❖ Equip the students to adapt to changing global and local needs through well designed curriculum and syllabus.
- ❖ Groom students to uphold professional ethics and develop leadership qualities
- ❖ Train students on issues related to social welfare

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO 1 - Successful professional career and/or higher studies by gaining knowledge in fundamental mathematics and biological principle.

PEO 2 - Provide strong foundation in the core biotechnology courses to evaluate real life problems and to propose biotechnological solutions with economical and social viability

PEO 3 - Sensitize on environmental, health and bioethical issues, Intellectual property rights, professional ethics and life-long learning through application orientated activities

PROGRAM OUTCOMES(POs)

The Biotechnology Graduates will have the ability to

PO 1: Engineering Knowledge

Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem Analysis

Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

PO3: Design/Development of Solutions

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct Investigations of Complex Problems

Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern Tool Usage

Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The Engineer and Society

Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and Sustainability

Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics

Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and Team Work

Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication

Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project Management and Finance

Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-Long Learning Skill

Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOME

PSO I :Apply the knowledge in core and allied fields to solve complex engineering problems.

PSO II : Able to expose their skills using latest tools to arrive cost effective and appropriate solutions.

PSO III : Apply the contextual knowledge with professional ethics to manage different projects in multi-disciplinary environment.

ABSTRACT

Fermentation is one of the oldest and most important biological processes used in food, beverage, and biotechnology industries. Among the microorganisms involved in fermentation, **yeast**, especially *Saccharomyces cerevisiae*, plays a vital role due to its ability to convert sugars into ethanol and carbon dioxide under anaerobic conditions. Yeast fermentation has wide applications in bread making, alcoholic beverage production, bioethanol manufacturing, and various industrial processes.

The present mini project report is prepared on the topic "**Yeast Fermentation**" following the academic requirements of a **II Year Biotechnology** student. The report provides a comprehensive understanding of the fermentation process, including the characteristics of yeast, the mechanism of fermentation, factors affecting yeast activity, industrial applications, and advantages of fermentation technology.

This report is designed as a structured academic document consisting of an introduction, literature review, methodology, observation format, results, discussion, and conclusion. Since this is a mini project, the report mainly focuses on the theoretical aspects of yeast fermentation while also providing a basic experimental procedure that can be performed in a laboratory. Observation tables and result formats are included for recording practical findings.

The study emphasizes the significance of yeast fermentation in biotechnology by highlighting its economic importance, eco-friendly nature, and contribution to sustainable industrial production. This report serves as a valuable learning resource for understanding microbial fermentation processes and their practical applications in modern biotechnology.

Keywords: Yeast fermentation, *Saccharomyces cerevisiae*, fermentation, ethanol production, biotechnology, industrial microbiology, anaerobic respiration.

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INTRODUCTION

1.1 Introduction to Fermentation

Fermentation is a natural biological process in which microorganisms such as bacteria, yeast, and fungi convert organic compounds, mainly carbohydrates, into simpler substances under anaerobic (oxygen-free) conditions. It is one of the oldest and most important biotechnological processes known to mankind. The process has been widely used for thousands of years in the production of food, beverages, medicines, and various industrial products. Today, fermentation is considered a fundamental technique in biotechnology because it provides an economical, sustainable, and environmentally friendly method for producing valuable biological products.

Among all microorganisms, ***Saccharomyces cerevisiae***, commonly known as baker's yeast, is the most widely used organism for fermentation. This unicellular eukaryotic fungus efficiently converts sugars such as glucose, fructose, sucrose, and maltose into ethanol and carbon dioxide through a series of enzymatic reactions. During fermentation, yeast obtains energy by breaking down glucose in the absence of oxygen. This metabolic pathway not only supports yeast growth but also produces commercially valuable products that are widely used in food processing and biofuel industries.

The overall chemical reaction for alcoholic fermentation can be represented as:



where:

- $\text{C}_6\text{H}_{12}\text{O}_6$ = Glucose
- $\text{C}_2\text{H}_5\text{OH}$ = Ethanol
- CO_2 = Carbon dioxide

This process is catalyzed by enzymes naturally present in yeast cells. The carbon dioxide released causes bread dough to rise during baking, while the ethanol produced is utilized in the preparation of alcoholic beverages and biofuels.

Fermentation is influenced by several environmental factors, including temperature, pH, sugar concentration, nutrient availability, and oxygen levels. The optimum temperature for ***Saccharomyces cerevisiae*** ranges from **25°C to 35°C**, while the ideal pH lies between **4.0**

and 6.0. Maintaining these conditions ensures maximum yeast activity and efficient ethanol production.

In recent years, there has been increasing interest in producing bioethanol from agricultural and food wastes because of growing environmental concerns and the depletion of fossil fuel resources. Agricultural wastes such as sugarcane bagasse, fruit peels, corn stalks, rice straw, and particularly **banana peel waste** are rich in carbohydrates and can serve as inexpensive substrates for fermentation.

Banana peel, which is usually discarded as waste, contains considerable amounts of cellulose, hemicellulose, starch, glucose, fructose, minerals, and other nutrients that support yeast growth. By converting banana peel waste into bioethanol through fermentation, environmental pollution can be reduced while simultaneously producing a renewable and eco-friendly source of energy. This approach also supports waste management and the principles of a circular bioeconomy.

Yeast fermentation plays an important role in many industrial sectors. In the food industry, it is used for the production of bread, beer, wine, vinegar, and fermented beverages. In the pharmaceutical industry, yeast is employed in the production of enzymes, vaccines, vitamins, antibiotics, and recombinant proteins. In the energy sector, fermentation is extensively used for bioethanol production, which serves as a renewable alternative to conventional fossil fuels.

Besides industrial applications, yeast fermentation has become an important research area in biotechnology. Scientists continuously study methods to improve fermentation efficiency by optimizing fermentation conditions, developing genetically improved yeast strains, and utilizing low-cost agricultural wastes as substrates. Such advancements contribute to sustainable development by reducing industrial waste, minimizing environmental pollution, and promoting renewable energy production.

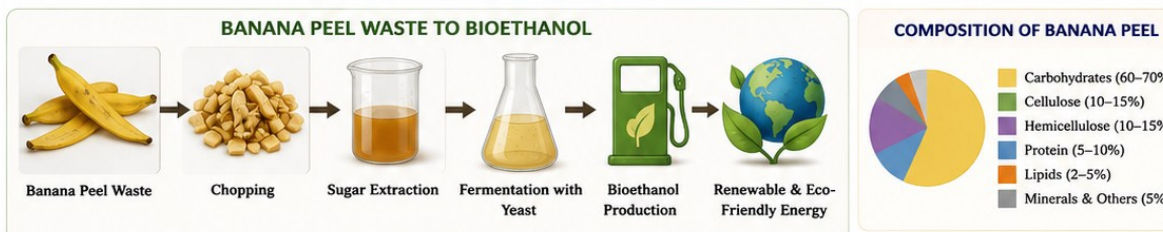
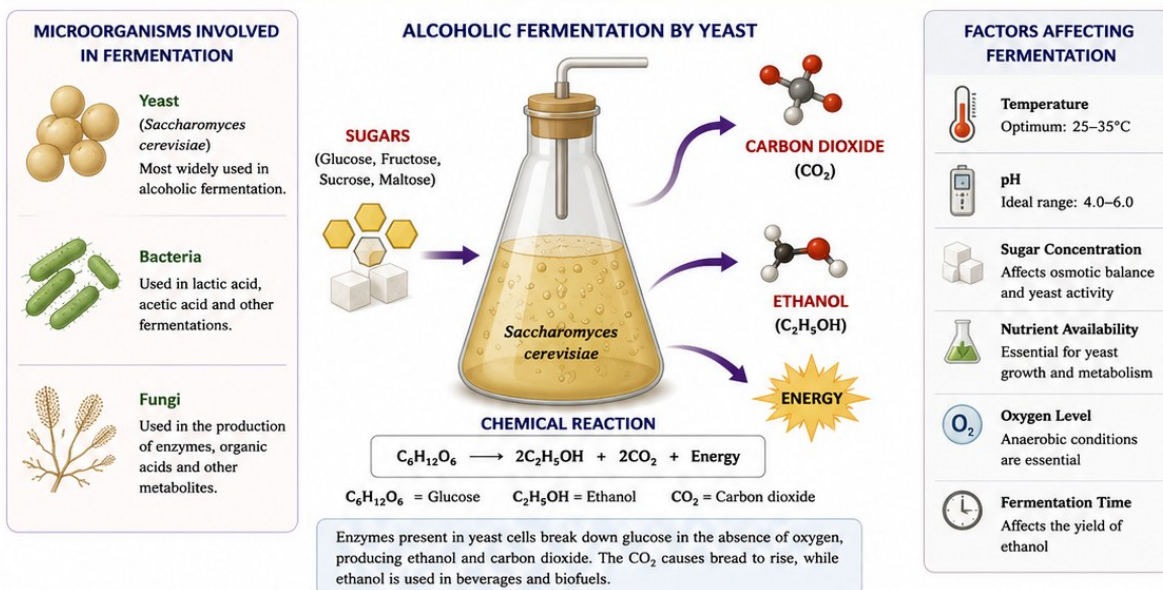
The present internship project focuses on the production of bioethanol using **banana peel waste** as the fermentation substrate and ***Saccharomyces cerevisiae*** as the fermenting microorganism. The study includes the preparation of banana peel extract, activation of yeast, preparation of fermentation medium, ethanol production, organic acid estimation, and cell viability analysis. This project demonstrates how biotechnology can convert agricultural waste into valuable products while promoting sustainable environmental practices.

Therefore, yeast fermentation is not only a simple biological process but also a powerful

biotechnological tool with significant applications in food production, pharmaceutical manufacturing, industrial biotechnology, environmental management, and renewable energy generation.

1.1 INTRODUCTION TO FERMENTATION

Fermentation is a natural biological process in which microorganisms convert organic compounds, mainly carbohydrates, into simpler substances under anaerobic (oxygen-free) conditions. It is one of the oldest and most important biotechnological processes known to mankind and is widely used in food, beverage, pharmaceutical and industrial production.



Thus, fermentation, especially by *Saccharomyces cerevisiae*, is a versatile and sustainable biotechnological process. It converts low-cost substrates such as banana peel waste into valuable products like bioethanol, contributing to waste management, environmental protection and renewable energy generation.

1.2 History of Yeast Fermentation

Fermentation is one of the oldest biotechnological processes practiced by humans and has played a significant role in the development of civilization. Long before the discovery of microorganisms, ancient people unknowingly used fermentation to preserve food and produce beverages. Historical evidence suggests that fermentation was first practiced around **7000–6000 BCE**, when early civilizations discovered that naturally occurring microorganisms could convert sugars into alcohol and other useful products.

The earliest known use of yeast fermentation dates back to **Ancient Egypt** around **3000 BCE**, where people used naturally occurring yeast to prepare bread. During bread making, they observed that the dough expanded after being left for some time. Although they did not understand the scientific reason, the carbon dioxide released by yeast during fermentation caused the dough to rise, producing soft and fluffy bread. Ancient Egyptians also used fermentation to prepare beer, which became one of their most important beverages.

Other ancient civilizations, including the **Mesopotamians, Babylonians, Greeks, Romans, Chinese, and Indians**, also practiced fermentation for producing bread, beer, wine, and fermented foods. Fermentation became an essential method for preserving food because it increased shelf life while improving flavor, aroma, texture, and nutritional value. These early applications established fermentation as an important part of agriculture and food processing.

During the **Middle Ages**, fermentation techniques continued to improve through practical experience. Monasteries in Europe became important centers for brewing beer and wine, where monks refined fermentation methods and maintained high-quality production standards. Although fermentation was widely practiced, people still believed it was a purely chemical process because microorganisms had not yet been discovered.

A major breakthrough occurred in the **17th century** when the Dutch scientist **Antonie van Leeuwenhoek** observed tiny living cells under a microscope for the first time. Among these cells were yeast cells, although their role in fermentation was not yet understood. His invention of the microscope opened the door for the study of microorganisms and laid the foundation for microbiology.

The scientific understanding of yeast fermentation greatly advanced during the **19th century** through the work of the French scientist **Louis Pasteur**. In **1857**, Pasteur demonstrated that fermentation is carried out by living microorganisms rather than occurring as a simple chemical reaction. He proved that yeast converts sugars into ethanol and carbon dioxide under

anaerobic conditions. His experiments established the field of **industrial microbiology** and earned him recognition as the "Father of Microbiology." Pasteur's discoveries revolutionized food preservation, brewing, and biotechnology.

With the advancement of science during the **20th century**, fermentation technology expanded rapidly. Scientists learned to isolate pure yeast cultures, optimize fermentation conditions, and improve product quality through controlled industrial processes. New fermentation equipment such as bioreactors and fermenters allowed large-scale production of ethanol, enzymes, antibiotics, vitamins, organic acids, and pharmaceuticals. Improvements in microbiology, genetics, and biochemical engineering further increased fermentation efficiency.

Today, yeast fermentation is one of the most important industrial biotechnological processes. The yeast ***Saccharomyces cerevisiae*** is widely used because of its rapid growth, high ethanol-producing capacity, safety, and ease of cultivation. It is extensively employed in the production of bread, beer, wine, bioethanol, baker's yeast, enzymes, vaccines, and several pharmaceutical products.

In recent years, the increasing demand for renewable energy has encouraged researchers to utilize agricultural wastes such as **banana peel, sugarcane bagasse, fruit peels, corn residues, rice straw, and food waste** as low-cost substrates for yeast fermentation. These renewable materials are converted into **bioethanol**, providing an environmentally friendly alternative to fossil fuels while reducing agricultural waste and environmental pollution.

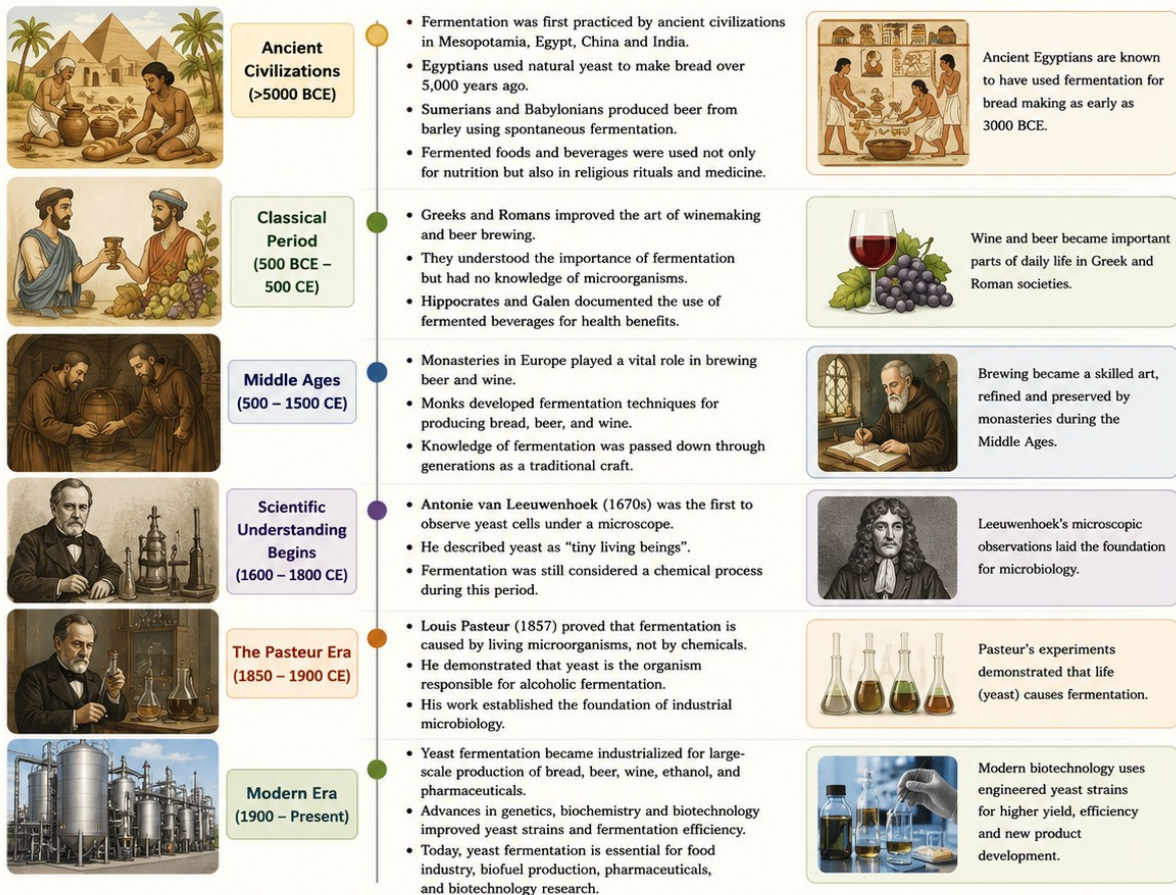
The present internship project follows this modern approach by utilizing **banana peel waste** as a carbohydrate-rich substrate and ***Saccharomyces cerevisiae*** for bioethanol production. This demonstrates how traditional fermentation technology has evolved into a sustainable and eco-friendly process with applications in renewable energy, waste management, and industrial biotechnology.

Thus, the history of yeast fermentation reflects the remarkable journey of biotechnology—from ancient bread making to modern biofuel production—and continues to play a vital role in scientific research, industrial development, and sustainable environmental practices.

1.2 HISTORY OF YEAST FERMENTATION

Fermentation is one of the earliest biotechnological processes practiced by humans. Long before the discovery of microorganisms, people used natural fermentation for the preservation and transformation of food and beverages. Yeast fermentation, in particular, has a rich historical background that has contributed significantly to the development of food industry, medicine, and biotechnology.

HISTORICAL TIMELINE

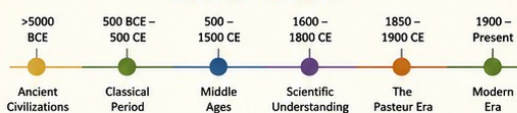


SIGNIFICANCE

- Yeast fermentation has evolved from an art to a science.
- It has played a crucial role in the development of human civilization.
- From traditional practices to advanced industrial applications, it continues to be a key technology in biotechnology.
- It supports food security, health, energy sustainability and economic growth.



SUMMARY TIMELINE



From ancient traditions to modern biotechnology, yeast fermentation has been a continuous journey of discovery, innovation and its applications for the benefit of humanity.

1.3 Introduction to *Saccharomyces cerevisiae*

Saccharomyces cerevisiae, commonly known as **baker's yeast**, is a unicellular eukaryotic fungus widely used in biotechnology, food processing, and industrial fermentation. It is one of the oldest and most extensively studied microorganisms due to its ability to efficiently convert sugars into ethanol and carbon dioxide through the process of alcoholic fermentation. Because of its simple growth requirements, rapid reproduction, and safe nature, *S. cerevisiae* has become the most preferred microorganism for laboratory research and industrial applications.

The genus **Saccharomyces** means "sugar fungus," reflecting its ability to metabolize various sugars such as glucose, fructose, sucrose, and maltose. During anaerobic conditions, yeast

breaks down these sugars to produce ethanol and carbon dioxide while releasing energy required for its growth and survival. Under aerobic conditions, however, yeast mainly undergoes respiration and produces more biomass instead of ethanol.

Saccharomyces cerevisiae is a single-celled organism with a round or oval shape, measuring approximately **5–10 µm** in diameter. It possesses a well-defined nucleus, cytoplasm, mitochondria, vacuoles, and a rigid cell wall composed mainly of glucans, mannoproteins, and chitin. Unlike bacteria, yeast is a eukaryotic microorganism, meaning its genetic material is enclosed within a membrane-bound nucleus.

The organism reproduces primarily by **budding**, where a small outgrowth develops on the parent cell, enlarges, and eventually separates to form a new daughter cell. Under favorable environmental conditions such as adequate nutrients, moisture, suitable temperature, and optimal pH, yeast multiplies rapidly, making it highly suitable for industrial fermentation processes.

The optimum growth temperature for *Saccharomyces cerevisiae* ranges between **25°C and 35°C**, while the optimum pH is between **4.0 and 6.0**. The yeast requires carbohydrates as its primary energy source and also needs nitrogen, minerals, vitamins, and trace elements for healthy growth. Environmental factors such as sugar concentration, oxygen availability, temperature, and pH significantly influence its fermentation efficiency.

One of the major advantages of *Saccharomyces cerevisiae* is its **Generally Recognized As Safe (GRAS)** status. It is non-pathogenic and has been safely used for centuries in food production. Because of its safety and high fermentation efficiency, it is extensively employed in the production of bread, beer, wine, bioethanol, enzymes, vaccines, pharmaceuticals, and recombinant proteins.

In industrial biotechnology, *Saccharomyces cerevisiae* plays an important role in producing **bioethanol**, a renewable fuel obtained by fermenting sugars present in agricultural wastes such as sugarcane bagasse, molasses, corn, fruit wastes, and banana peels. During the fermentation process, the yeast converts fermentable sugars into ethanol, which can be used as an alternative to fossil fuels. This not only supports renewable energy production but also helps reduce environmental pollution by utilizing agricultural waste materials.

In the present internship project, *Saccharomyces cerevisiae* was selected as the fermenting microorganism because of its high ethanol-producing ability, rapid growth rate, easy cultivation, and cost-effectiveness. The yeast was activated before fermentation and

inoculated into a banana peel extract medium enriched with nutrients. During incubation, the yeast metabolized the sugars present in the banana peel extract and produced ethanol and carbon dioxide. The fermentation process was later evaluated through ethanol estimation, organic acid production, and cell viability tests.

Due to its outstanding biological characteristics, industrial importance, and versatility, *Saccharomyces cerevisiae* remains one of the most valuable microorganisms in biotechnology. Its extensive applications in food, pharmaceuticals, renewable energy, and environmental biotechnology continue to make it a preferred organism for research and industrial-scale fermentation.

1.3 INTRODUCTION TO *Saccharomyces cerevisiae*

Saccharomyces cerevisiae is a unicellular, eukaryotic fungus belonging to the kingdom Fungi and phylum *Ascomycota*. It is commonly known as baker's yeast, brewer's yeast or simply yeast. It is one of the most widely studied and commercially important microorganisms because of its ability to ferment sugars efficiently and produce ethanol and carbon dioxide. It has been used by humans for thousands of years in the production of bread, alcoholic beverages and other fermented foods.

S. cerevisiae is non-pathogenic, safe (GRAS – Generally Recognized As Safe) and easy to cultivate in the laboratory as well as in industry. It grows rapidly under aerobic conditions and ferments sugars under anaerobic conditions. The organism can utilize a wide range of simple and complex sugars such as glucose, fructose, sucrose, maltose and galactose.

The cells of *S. cerevisiae* are typically oval or spherical in shape, measuring about 5–10 µm in diameter. It reproduces asexually by budding, where a small outgrowth (bud) appears on the parent cell, develops into a new cell and finally separates. Under unfavorable conditions, it can form ascospores through sexual reproduction.

It grows well at a temperature range of 25–35°C, with an optimum at 30°C and pH range of 4.0–6.0. It requires nutrients such as carbon sources, nitrogen sources, vitamins, minerals and trace elements for healthy growth and high fermentation efficiency.

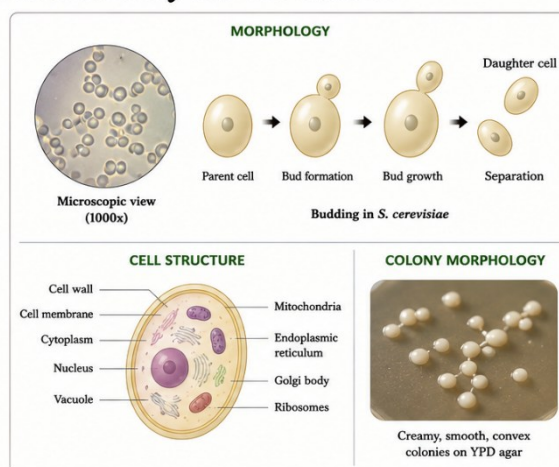
S. cerevisiae plays a vital role in biotechnology and is used in the production of bread, beer, wine, bioethanol, enzymes, vaccines, vitamins and pharmaceuticals. In the present study, this yeast is used as the fermenting microorganism for the production of bioethanol from banana peel waste.

KEY CHARACTERISTICS

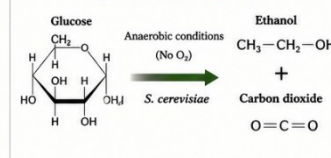
- Unicellular, eukaryotic fungus.
- Reproduces asexually by budding.
- Non-pathogenic and safe for industrial use.
- Grows well at 25–35°C and pH 4.0–6.0.
- Utilizes a wide range of sugars.
- Produces ethanol and carbon dioxide during fermentation.
- Easy to culture, maintain and scale up.
- Extensively used in food, beverage, pharmaceutical and biofuel industries.

TAXONOMY

Kingdom	: Fungi
Phylum	: Ascomycota
Class	: Saccharomycetes
Order	: Saccharomycetales
Family	: Saccharomycetaceae
Genus	: <i>Saccharomyces</i>
Species	: <i>Saccharomyces cerevisiae</i>



FERMENTATION ABILITY



IMPORTANCE

- Used in baking (CO₂ production).
- Used in brewing and winemaking (ethanol production).
- Used in bioethanol production.
- Source of enzymes, vitamins and pharmaceutical products.
- Model organism in biological research.

Saccharomyces cerevisiae is a versatile and industrially important yeast with excellent fermentative ability, making it the preferred microorganism for bioethanol production and various biotechnological applications.

1.4 Banana Peel as a Substrate for Fermentation

Banana is one of the most widely consumed fruits in the world, and its peel accounts for approximately 30–40% of the total fruit weight. Every year, large quantities of banana peel are discarded as agricultural and household waste, leading to environmental pollution and waste management challenges. However, banana peel is rich in carbohydrates, cellulose, hemicellulose, starch, pectin, minerals, and small amounts of proteins, making it a valuable substrate for microbial fermentation.

The carbohydrate content present in banana peel can be converted into fermentable sugars

through suitable processing methods. These sugars serve as an excellent carbon source for the growth of *Saccharomyces cerevisiae*. During fermentation, the yeast utilizes these sugars and converts them into ethanol and carbon dioxide through enzymatic reactions.

Using banana peel as a fermentation substrate offers several advantages. It is inexpensive, readily available, renewable, biodegradable, and helps in reducing environmental pollution caused by food waste. The conversion of banana peel into bioethanol also supports sustainable waste management and promotes the circular bioeconomy by transforming waste into a valuable renewable energy source.

In this internship project, banana peel was selected as the fermentation substrate because of its high carbohydrate content and easy availability. The peel was cleaned, chopped, blended, and processed to obtain a sugar-rich extract, which was then used as the fermentation medium for ethanol production.

1.5 Principle of Ethanol Production by Yeast Fermentation

Ethanol production by yeast fermentation is based on the ability of *Saccharomyces cerevisiae* to convert sugars into ethanol and carbon dioxide under anaerobic conditions. During this biological process, yeast cells utilize glucose as their primary energy source. Through a series of enzyme-catalyzed reactions, glucose is converted into pyruvate via glycolysis. In the absence of oxygen, pyruvate is further converted into ethanol and carbon dioxide.

The overall reaction is represented as:



Where:

- $\text{C}_6\text{H}_{12}\text{O}_6$ = Glucose
- $\text{C}_2\text{H}_5\text{OH}$ = Ethanol
- CO_2 = Carbon dioxide

Several factors influence the efficiency of ethanol production, including temperature, pH, sugar concentration, yeast concentration, nutrient availability, and fermentation time. The optimum temperature for yeast fermentation is between **30°C and 35°C**, while the optimum pH ranges from **4.5 to 5.5**.

In this study, activated yeast was inoculated into a banana peel extract containing sufficient nutrients. The fermentation process was carried out under controlled conditions, allowing yeast to convert the available sugars into ethanol efficiently.

1.6 Industrial Applications of Ethanol Fermentation

Ethanol fermentation has numerous industrial applications due to its economic importance and environmentally friendly nature. It is widely used in food, beverage, pharmaceutical, biotechnology, and renewable energy industries.

The major applications include:

- Production of alcoholic beverages such as beer, wine, and spirits.
- Bread making, where carbon dioxide produced by yeast causes dough to rise.
- Bioethanol production for use as renewable transportation fuel.
- Pharmaceutical industries for manufacturing disinfectants, solvents, and medicinal products.
- Cosmetic industries for perfumes, lotions, and sanitizers.
- Chemical industries for producing organic solvents and industrial chemicals.
- Biotechnology laboratories for research and microbial studies.

The increasing demand for renewable fuels has significantly increased the importance of ethanol fermentation. Bioethanol reduces greenhouse gas emissions and provides an eco-friendly alternative to conventional fossil fuels.

1.7 Problem Statement

The rapid increase in agricultural and food waste has become a major environmental concern worldwide. Banana peel is one of the most commonly discarded organic wastes despite containing a significant amount of fermentable carbohydrates. Improper disposal of banana peel contributes to environmental pollution and increases waste management costs.

At the same time, the depletion of fossil fuel resources and increasing energy demands have created a need for sustainable renewable energy sources. Bioethanol produced through microbial fermentation offers an environmentally friendly alternative to conventional fuels.

Therefore, this study aims to utilize banana peel waste as an economical substrate for

bioethanol production using *Saccharomyces cerevisiae*, thereby reducing waste while producing a valuable renewable energy resource.

1.8 Aim of the Study

To produce bioethanol from banana peel waste using *Saccharomyces cerevisiae* through the fermentation process and to evaluate its ethanol-producing efficiency under laboratory conditions.

1.9 Objectives

The objectives of this study are:

- To understand the principle of yeast fermentation.
- To utilize banana peel as a low-cost fermentation substrate.
- To prepare and activate *Saccharomyces cerevisiae* culture.
- To produce ethanol through anaerobic fermentation.
- To estimate ethanol production.
- To analyze organic acid production.
- To study yeast cell viability after fermentation.
- To understand the industrial applications of bioethanol production.

1.10 Scope of the Study

The present study focuses on the laboratory-scale production of bioethanol from banana peel waste using *Saccharomyces cerevisiae*. It includes banana peel processing, yeast activation, preparation of fermentation medium, fermentation, ethanol estimation, organic acid analysis, and cell viability testing.

The knowledge gained from this study can be applied in renewable energy production, industrial biotechnology, environmental biotechnology, and sustainable waste management. The study also demonstrates the potential of agricultural waste as an alternative feedstock for bioethanol production.

1.11 Significance of the Study

The production of bioethanol from banana peel waste is an environmentally friendly and sustainable approach to waste utilization. It helps reduce agricultural waste while producing a valuable renewable fuel. This process minimizes dependence on fossil fuels and contributes to cleaner energy production.

From a biotechnological perspective, the study enhances the understanding of microbial fermentation, enzyme activity, and industrial applications of *Saccharomyces cerevisiae*. It also provides practical laboratory experience in fermentation technology, ethanol estimation, organic acid analysis, and microbial handling.

The findings of this project may contribute to future research aimed at improving bioethanol production from low-cost agricultural wastes. The study supports sustainable development, waste valorization, and the advancement of renewable energy technologies.

REVIEW OF LITERATURE

2.1 Overview of Yeast Fermentation

Yeast fermentation is a biological process in which yeast converts sugars into ethanol and carbon dioxide under anaerobic conditions. This process is one of the oldest applications of biotechnology and has been used for thousands of years in food production, brewing, and winemaking. Among various yeast species, *Saccharomyces cerevisiae* is the most commonly used because of its high fermentation efficiency, rapid growth, and ability to produce large amounts of ethanol.

During fermentation, yeast utilizes glucose as its primary energy source. The glucose is metabolized through glycolysis, producing pyruvate, which is further converted into ethanol and carbon dioxide. This process allows yeast to survive in oxygen-limited environments while generating energy for growth and reproduction.

Yeast fermentation is economically important because it produces products such as bread, beer, wine, industrial alcohol, bioethanol, enzymes, organic acids, and pharmaceutical compounds. In recent years, fermentation technology has gained importance due to its role in renewable energy production through bioethanol.

2.2 Biology and Morphology of *Saccharomyces cerevisiae*

Saccharomyces cerevisiae is a unicellular eukaryotic fungus commonly referred to as baker's

yeast. It belongs to the Kingdom Fungi and reproduces mainly through budding. The cells are oval or spherical and measure approximately **5–10 μm** in diameter.

The yeast cell consists of a cell wall, plasma membrane, cytoplasm, nucleus, mitochondria, vacuoles, ribosomes, and other organelles. The cell wall is composed mainly of glucans, mannoproteins, and chitin, which provide strength and protection.

Yeast grows rapidly under favorable environmental conditions. The optimum temperature ranges from **30–35°C**, while the optimum pH lies between **4.5 and 5.5**. Because of its rapid growth, high ethanol production, and GRAS (Generally Recognized As Safe) status, *S. cerevisiae* is widely used in food industries, pharmaceuticals, biotechnology research, and biofuel production.

2.3 Composition and Importance of Banana Peel

Banana peel is an abundant agricultural waste generated from domestic consumption and food industries. It accounts for approximately **30–40%** of the total weight of the fruit and is usually discarded. However, banana peel is rich in carbohydrates, cellulose, hemicellulose, starch, pectin, dietary fiber, proteins, minerals, vitamins, and antioxidants.

The carbohydrate-rich composition of banana peel makes it a suitable substrate for microbial fermentation. The sugars present in banana peel serve as an excellent carbon source for *Saccharomyces cerevisiae*. Using banana peel for bioethanol production reduces environmental pollution, minimizes agricultural waste, and provides a low-cost raw material for renewable energy production.

The utilization of banana peel also supports sustainable waste management by converting biodegradable waste into valuable biofuel.

2.4 Biochemical Pathway of Ethanol Fermentation

The biochemical pathway of ethanol fermentation begins with the breakdown of glucose through glycolysis. During glycolysis, one molecule of glucose is converted into two molecules of pyruvate while producing ATP and NADH.

Under anaerobic conditions, pyruvate is converted into acetaldehyde by the enzyme pyruvate

decarboxylase. Acetaldehyde is then reduced to ethanol by alcohol dehydrogenase using NADH.

The overall reaction is:



This process enables yeast to generate energy without oxygen while producing ethanol and carbon dioxide as end products.

2.5 Factors Affecting Ethanol Production

Several environmental and nutritional factors influence ethanol production by yeast.

The important factors include:

- Temperature
- pH
- Sugar concentration
- Yeast concentration
- Fermentation time
- Oxygen availability
- Nutrient supplementation
- Initial substrate concentration

The optimum temperature for fermentation is **30–35°C**, while the optimum pH ranges from **4.5 to 5.5**. Excessive temperature or unsuitable pH reduces yeast growth and ethanol yield.

Proper nutrient supplementation using nitrogen and mineral sources also enhances fermentation efficiency.

2.6 Industrial Applications of Bioethanol

Bioethanol has become one of the most important renewable fuels in modern industries. It is

widely used as an alternative fuel because it burns cleanly and reduces greenhouse gas emissions.

Major applications include:

- Transportation fuel
- Fuel blending with petrol
- Alcoholic beverage production
- Pharmaceutical industries
- Cosmetic industries
- Chemical manufacturing
- Laboratory solvents
- Food processing industries

The use of bioethanol contributes to sustainable development by reducing dependence on fossil fuels.

2.7 Recent Advances in Bioethanol Production

Recent developments in biotechnology have significantly improved bioethanol production. Researchers have developed genetically engineered yeast strains with enhanced fermentation efficiency and stress tolerance.

Modern techniques include:

- Genetic engineering of yeast
- Simultaneous saccharification and fermentation (SSF)
- Cell immobilization
- Continuous fermentation
- Use of lignocellulosic biomass
- Nanotechnology-based fermentation monitoring

- Bioreactor optimization

Agricultural wastes such as banana peel, sugarcane bagasse, rice straw, fruit peels, and corn residues are now extensively used as inexpensive feedstocks for bioethanol production.

These advances improve ethanol yield while reducing production costs and environmental pollution.

2.8 Research Gap

Although numerous studies have investigated bioethanol production from agricultural wastes, there remains a need to improve fermentation efficiency, reduce production costs, and utilize locally available waste materials more effectively.

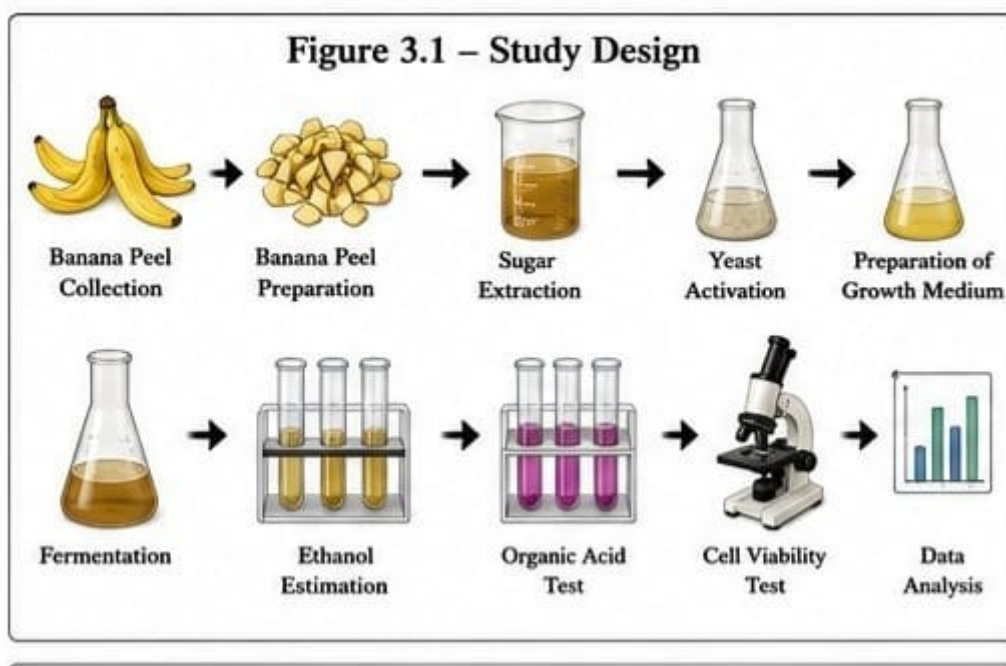
Banana peel is produced in large quantities and often discarded without proper utilization. Limited studies have focused on optimizing ethanol production from banana peel under laboratory conditions using *Saccharomyces cerevisiae*.

The present internship study attempts to utilize banana peel waste as a low-cost fermentation substrate for bioethanol production. It also evaluates ethanol production, organic acid formation, and yeast cell viability, thereby contributing to sustainable waste management and renewable energy research.

MATERIALS AND METHODS

3.1 Study Design

The present study was designed to evaluate the production of bioethanol from banana peel waste using *Saccharomyces cerevisiae*. The experiment involved collecting fresh banana peels, extracting fermentable sugars, activating the yeast culture, preparing a nutrient-rich fermentation medium, and carrying out fermentation under controlled laboratory conditions. The fermented sample was then analyzed for ethanol production, organic acid content, and yeast cell viability. The overall workflow of the experiment is shown in the study design flowchart.



3.2 Materials and Equipment

Chemicals

- Baker's yeast (*Saccharomyces cerevisiae*)
- Glucose
- Ammonium sulfate ($(\text{NH}_4)_2\text{SO}_4$)
- Sodium dihydrogen phosphate (NaH_2PO_4)
- Magnesium sulfate (MgSO_4)
- Distilled water
- Phenolphthalein indicator
- 0.1 N Sodium hydroxide (NaOH)
- Ethanol test reagent (Green colour indicator)

Equipment

- Beaker
- Conical flask
- Measuring cylinder

- Weighing balance
- Hot plate
- Blender
- Centrifuge
- Incubator
- Microscope
- Pipettes
- Test tubes
- Glass rod

3.3 Collection and Preparation of Banana Peel

Fresh banana peels were collected from locally available ripe bananas. The peels were thoroughly washed with distilled water to remove dust and impurities. They were cut into small pieces using a sterile knife and blended into a fine paste. The paste was heated gently to facilitate sugar extraction and then filtered to obtain a clear liquid extract. This extract served as the substrate for ethanol fermentation.

3.4 Preparation of Sugar Extract

The banana peel paste was mixed with distilled water and heated to extract soluble sugars. Additional glucose was added to improve the fermentable sugar concentration. The mixture was boiled for approximately 15–20 minutes and then centrifuged at **3000 rpm for 10 minutes**. The supernatant containing fermentable sugars was collected carefully and stored under sterile conditions until further use.

3.5 Activation of Yeast Culture

Approximately **5 g of baker's yeast** was weighed accurately and transferred into a sterile beaker containing **10 mL of lukewarm distilled water**. The mixture was stirred gently and allowed to stand for **15–20 minutes** at room temperature to activate the yeast cells. Activated yeast showed visible frothing, indicating active metabolism and readiness for fermentation.

3.6 Preparation of Growth Medium

A nutrient-rich growth medium was prepared using **50 mL of distilled water** in a sterile conical flask. The following nutrients were added:

- Glucose
- Ammonium sulfate
- Sodium dihydrogen phosphate
- Magnesium sulfate

The solution was mixed thoroughly until all chemicals dissolved completely. The medium was sterilized to prevent contamination and allowed to cool before inoculation with activated yeast.

3.7 Fermentation Procedure

The prepared sugar extract was mixed with the nutrient medium and inoculated with activated yeast culture. The fermentation flask was covered with a cotton plug to minimize contamination while allowing gas exchange. The flask was incubated at **30–35°C** for **48–72 hours** under anaerobic conditions.

During fermentation, yeast converted the sugars present in the banana peel extract into ethanol and carbon dioxide. Gas bubble formation, frothing, and changes in odor were recorded as indicators of active fermentation.

3.8 Ethanol Estimation Test

After completion of fermentation, ethanol production was estimated using a green colour indicator reagent. A measured volume of the fermented sample was mixed with the reagent, and the colour intensity was observed. The ethanol concentration was calculated using the standard formula provided in the laboratory manual. The final ethanol concentration obtained during the experiment was **0.609**.

3.9 Organic Acid Production Test

To determine organic acid production, **5 mL** of the fermented sample was centrifuged to obtain a clear supernatant. The supernatant was titrated against **0.1 N NaOH** using **phenolphthalein** as an indicator. The endpoint was indicated by the appearance of a light

pink colour. The acidity was calculated using the standard formula, and the organic acid concentration obtained was **0.512**.

3.10 Cell Viability Test

The viability of yeast cells after fermentation was examined using a compound microscope. A small drop of the fermented culture was placed on a clean glass slide, covered with a coverslip, and observed under **40× magnification**. Healthy yeast cells appeared oval in shape, while budding cells indicated active cell division and successful fermentation.

3.11 Statistical Analysis

The experimental observations were recorded carefully and arranged in tabular form. The values obtained for ethanol production and organic acid concentration were compared and interpreted using simple statistical methods such as mean and percentage where applicable. The results were represented using graphs and tables for better understanding.

3.12 Laboratory Safety

The experiment was carried out by following standard laboratory safety guidelines. A laboratory coat, gloves, and safety mask were worn throughout the experiment. All glassware and equipment were sterilized before use to prevent contamination. Chemicals were handled carefully according to laboratory protocols, and biological waste was disposed of safely following institutional biosafety guidelines. Proper hand hygiene was maintained before and after performing the experiment.

3.13 Limitations

The study was conducted on a laboratory scale and therefore may not represent industrial-scale ethanol production. Variations in banana peel composition, sugar content, fermentation temperature, and yeast activity could influence ethanol yield. Time constraints and limited laboratory resources also affected the optimization of fermentation conditions. Despite these limitations, the study successfully demonstrated the potential of banana peel waste as a substrate for bioethanol production using *Saccharomyces cerevisiae*.

RESULTS AND DISCUSSION

4.1 Observation during Yeast Activation

The baker's yeast (*Saccharomyces cerevisiae*) was activated by mixing 5 g of dry yeast with 10 mL of lukewarm distilled water. After approximately 15–20 minutes, visible frothing and bubble formation were observed, indicating that the yeast cells had become metabolically active. The activated culture appeared creamy and showed signs of vigorous respiration, confirming its suitability for fermentation. Healthy yeast activation is essential because it directly influences ethanol production efficiency.

4.2 Banana Peel Sugar Extraction

Fresh banana peels were washed, chopped into small pieces, blended, and heated with distilled water to extract fermentable sugars. The mixture was centrifuged to separate the solid residue from the liquid extract. The obtained sugar extract appeared light brown with a slightly sweet smell, indicating the presence of soluble carbohydrates suitable for yeast fermentation. This extract served as the primary carbon source during ethanol production.

4.3 Fermentation Observations

The activated yeast culture was inoculated into the prepared banana peel sugar extract containing the nutrient medium. The fermentation flask was incubated at **30–35°C** under anaerobic conditions.

During fermentation, the following observations were recorded:

- Formation of gas bubbles due to carbon dioxide release.
- Frothing on the surface of the medium.
- Slight alcoholic odor indicating ethanol production.
- Increased turbidity due to yeast growth.
- No contamination observed throughout the incubation period.

These observations confirmed that the yeast was actively fermenting the sugars present in the banana peel extract.

Table 4.1: Fermentation observations.

Observation	Result
Gas bubbles	Present
Froth formation	Present
Alcoholic smell	Present
Contamination	Not observed
Yeast growth	Good

4.4 Ethanol Production Results

Following fermentation, ethanol production was estimated using the laboratory ethanol test. The fermented sample developed a green colour with the ethanol indicator reagent, confirming ethanol formation. The ethanol concentration was calculated using the prescribed formula.

The experimental ethanol concentration obtained was:

Ethanol concentration = 0.609

This result demonstrated that banana peel contains sufficient fermentable sugars to support ethanol production by *Saccharomyces cerevisiae*.

Table 4.2: Ethanol production results.

Parameter	Observation
Colour developed	Green
Ethanol concentration	0.609

4.5 Organic Acid Production Results

Organic acid production was determined by titrating the fermented sample with **0.1 N NaOH** using phenolphthalein indicator. A light pink endpoint indicated successful neutralization.

The acidity was calculated using the standard laboratory equation.

The calculated acidity was:

Organic acid concentration = 0.512

The presence of organic acids indicates normal yeast metabolism during fermentation.

Table 4.3: Organic acid production.

Parameter	Observation
Indicator	Phenolphthalein
Endpoint	Light pink

Organic acid concentration **0.512**

4.6 Cell Viability Results

The viability of yeast cells was examined microscopically after fermentation. Numerous healthy oval-shaped cells with budding structures were observed. Only a few inactive cells were present, indicating that the fermentation conditions successfully supported yeast survival and growth throughout the experiment.

These findings confirmed that the yeast culture remained viable and metabolically active until the completion of fermentation.

Table 4.4: Cell viability observations.

Parameter	Observation
Cell shape	Oval
Budding cells	Present
Cell viability	High

4.7 Discussion

The present internship study successfully demonstrated the production of bioethanol from banana peel waste using *Saccharomyces cerevisiae*. Banana peel proved to be an effective substrate because it contains fermentable carbohydrates that support yeast growth and ethanol production.

The activation of yeast resulted in rapid cell growth, which enhanced fermentation efficiency. Continuous gas production and froth formation confirmed active metabolism throughout the fermentation period. The ethanol concentration obtained (**0.609**) indicated successful conversion of sugars into ethanol, while the measured organic acid concentration (**0.512**) reflected normal microbial metabolic activity.

Microscopic examination revealed healthy budding yeast cells, confirming that the fermentation conditions maintained high cell viability. Overall, the experiment demonstrated that banana peel waste can be effectively utilized as a low-cost substrate for bioethanol production, supporting sustainable waste management and renewable energy generation.

4.8 Statistical Summary

The experimental observations were summarized and interpreted using simple statistical methods.

Table 4.5: Statistical summary.

Parameter	Result
Yeast activation	Successful
Fermentation	Successful
Ethanol concentration	0.609
Organic acid concentration	0.512
Cell viability	High

The results indicate that the experimental procedure was successful and produced consistent observations throughout the study.

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The present internship project successfully demonstrated the production of bioethanol from banana peel waste using *Saccharomyces cerevisiae*. The yeast culture was effectively activated and utilized the sugars extracted from banana peel to produce ethanol under anaerobic conditions.

The experiment confirmed that banana peel is an inexpensive and renewable substrate suitable for ethanol production. The measured ethanol concentration (**0.609**) and organic acid concentration (**0.512**) indicated successful fermentation. The cell viability test further confirmed that the yeast remained healthy and active throughout the fermentation process.

Overall, this study highlights the potential of converting agricultural waste into valuable biofuel while reducing environmental pollution. The project also provided practical knowledge of fermentation technology, microbial handling, ethanol estimation, organic acid analysis, and laboratory techniques.

5.2 Future Scope

The present study provides a foundation for future research in bioethanol production.

Future improvements may include:

- Optimization of fermentation conditions to improve ethanol yield.
- Use of genetically improved yeast strains for higher productivity.
- Utilization of other agricultural wastes such as sugarcane bagasse, rice straw, corn stalks, and fruit wastes.
- Scale-up of the fermentation process for industrial bioethanol production.
- Application of modern bioreactors and continuous fermentation systems.
- Investigation of cost-effective downstream processing techniques.
- Development of sustainable waste-to-energy technologies for commercial applications.

The continued advancement of bioethanol production technology will contribute significantly to renewable energy generation and environmental sustainability

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